Harness Engineering

General Agents, Chapter 9/12 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/harness-engineering

EN: Agent = Model + Harness. The model gives intelligence; the harness makes it reliable. 5 verbs: Constrain, Inform, Verify, Correct, Escalate. 'A guardrail always lives in the harness, never in the prompt.'
Roman Urdu: Agent = Model + Harness. Model intelligence deta, harness usay reliable banata hai. 5 verbs: Constrain, Inform, Verify, Correct, Escalate. 'Guardrail hamesha harness mein rehta hai, prompt mein kabhi nahi.'

1. Constrain + Inform

- **Permission rules:** Allow/Ask/Deny, sorted by BLAST RADIUS not frequency. Priority: Deny > Ask > Allow. Deny patterns match command TEXT, not intent -- they're tripwires, not a wall.
- **Sandboxes** don't trust the agent -- if something breaks, it stays contained (worktree, filesystem, network, branch fences).
- **Context surfaces:** rules file (always true), skills (how to do specific tasks), connectors (what it can reach). AX test: 'could a competent stranger take the right next step from this text alone?'

2. Verify + Correct

- Hooks are self-running verification with the power to refuse: PreToolUse/Stop can BLOCK (exit 2); PostToolUse can't undo, only feeds back to the next turn.
- Typed output beats free-text verdicts -- a malformed verdict escalates, it never gets guessed.
- **2 clocks:** Recovery (fast -- retry/skip/rollback by failure type) and Ratchet (slow -- change the harness so the mistake becomes impossible). 4 failure classes: Context, Constraint, Verification, Planning.

3. Limits + Staying the Engineer

- **3 forces limiting a harness:** Capability vs Control trade-off (max-tight = min-ambition), Harness Coupling (couple to contracts, not behaviors), Rule Debt (every rule costs -- review monthly).
- 0.95 reliability over 20 steps = ~36% clean completion -- harness changes alone have shown up to 10x gains.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Agent = Model + Harness	Model gives intelligence; harness makes it reliable.	Model intelligence deta; harness reliable banata.
Blast radius	How much damage a permission could cause -- the sort key for permission rules.	Permission se kitna damage ho sakta -- permission rules ka sort key.
Guardrail in harness, not prompt	The core rule: enforcement lives in code/config, never relies on the model following instructions.	Core rule: enforcement code/config mein rehta, model ke instruction follow karne par depend nahi karta.
Maker-checker (typed)	A JSON-schema verdict, validated field by field, instead of a free-text impression.	JSON-schema verdict, field-by-field validated, free-text impression ki jagah.
Ratchet	Change the harness so a caught mistake becomes structurally impossible to repeat.	Harness badlo taake pakri gayi galti dobara structurally mumkin na rahe.

Quick Self-Check

Question	Answer
Why must a guardrail live in the harness, never the prompt?	A prompt instruction can be ignored or overridden by the model; harness-level code/config enforces regardless of what the model 'decides'.
What's the difference between the Recovery clock and the Ratchet clock?	Recovery is fast (retry/skip/rollback a single failure); Ratchet is slow (permanently change the harness so that failure class can't recur).
Why are deny-list patterns called 'tripwires', not a wall?	They match command TEXT, not intent -- a rephrased dangerous command can slip past a text-based deny rule.
What happens to full-chain reliability at 0.95 per-step over 20 steps?	Only about 36% of runs finish cleanly -- small per-step failure rates compound badly over long chains.

Where This Leads / Sources

Leads to: Graph Engineering, Trusting the Checker. Full 8 practice-project harness builds in 07; 18-Q quiz in 10-test-your-understanding.

Source: docs/harness-engineering/ (this repo), content originally via Zia Tutor AI connector.